AP CALCULUS AB – UNIT 4

Contextual Applications of Differentiation

Expanded Complete-Coverage Coloured Review

Name: ____________ **Class:** ____________ **Date:** ____________

Review Scope and Learning Objectives

- Interpret derivatives and higher derivatives with correct units and contextual meaning.
- Analyze position, velocity, acceleration, speed, direction, and turning behavior.
- Model applied rates and solve related-rates problems from geometric or physical constraints.
- Use linearization and differentials for approximation and error analysis.
- Recognize valid indeterminate forms and apply L'Hopital's Rule appropriately.
- Review every idea, theorem hypothesis, representation, procedure, and common misconception appearing in the website lessons.

Complete Lesson Coverage Audit

Every website lesson is given its own mastery section below. Each section contains the complete lesson objectives, strategic reminders, a solved model, and foundation-to-challenge practice. The final synthesis sections mix lessons so that students must select procedures rather than follow a stated template.

Complete Website Lesson Coverage Map

Lesson	Website title	Ideas explicitly reviewed
4.1	Interpretations of the Derivative	Interpret a derivative value as an instantaneous rate of change in context.; State derivative units by dividing output units by input units.; Use the sign and magnitude of a derivative to describe local behavior.; Distinguish the meaning of a quantity, its first derivative, and its second derivative.
4.2	Straight Line Motion	Relate position, velocity, speed, and acceleration using derivatives.; Determine direction of motion, rest times, and changes in speed.; Analyze motion from formulas, graphs, and tables.; Calculate displacement and total distance over intervals using turning times.
4.3	Other Rates of Change	Calculate and interpret derivatives in scientific, economic, and geometric contexts.; Distinguish a modeled quantity from its instantaneous rate of change.; Use correct units and signs in applied interpretations.; Combine derivative rules to calculate rates for derived quantities.
4.4	Related Rates	Identify changing quantities linked by a geometric or physical constraint.; Differentiate a constraint with respect to time using implicit differentiation.; Substitute information from a specified instant and solve for an unknown rate.; Interpret the sign and units of the resulting rate.
4.5	Linear Approximation	Construct the linearization of a differentiable function at a convenient input.; Approximate nearby values using a tangent line.; Use differentials to estimate changes and propagated error.; Use concavity to determine whether a linear approximation overestimates or underestimates.
4.6	L'Hopital's Rule	Recognize the indeterminate quotient forms $0/0$ and ∞/∞.; Verify the hypotheses of L'Hopital's Rule before differentiating numerator and denominator.; Evaluate indeterminate limits, including repeated applications when justified.; Rewrite selected products, differences, and powers into allowed quotient forms.

Unit Knowledge, Definitions, and Theorem Library

Key Knowledge, Formulas, and Theorems

- In context, $Q'(a)$ describes the instantaneous change of Q at input a and has units output-units per input-unit; $Q''(a)$ has squared input units in the denominator.
- Motion relations: $v = s'$, $a = v' = s''$, speed $= |v|$. Direction depends on the sign of velocity; speeding up occurs when v and a have the same sign.
- Displacement is final position minus initial position. Total distance requires splitting at every actual direction change and adding absolute position changes.
- Applied derivatives can represent population, temperature, area, volume, cost, revenue, density, or other rates; every interpretation must include sign, instant, and units.
- Related-rates workflow: label a model, write a constraint, differentiate with respect to time, substitute the specified instant, solve, and interpret.
- Linearization at a: $L(x) = f(a) + f'(a)(x - a)$. Differentials give $\Delta y \approx dy = f'(a)\,dx$.
- Concavity determines the local direction of linearization error: concave up gives tangent underestimates; concave down gives tangent overestimates.
- L'Hopital's Rule applies only after verifying $0/0$ or ∞/∞. Differentiate numerator and denominator separately and reevaluate.
- Products, differences, and powers with indeterminate forms may be rewritten as quotients or handled by logarithms before applying the rule.
- Forms such as $1/0$, $0/\infty$, and $\infty/0$ are not indeterminate and do not justify L'Hopital.

Strategy and Precision Checklist

1. Name the representation and the requested quantity before calculating.
2. State all hypotheses before using a theorem.
3. Preserve exact values in non-calculator work and use radians for trigonometric calculus.
4. A graph or table supports an estimate; an analytic proof requires a definition, theorem, or valid algebraic argument.
5. Contextual conclusions must include the instant, sign, units, and meaning.
6. Candidate conditions such as a zero derivative, zero second derivative, or zero denominator require sign/domain verification before a conclusion.

Lesson 4.1: Interpretations of the Derivative

Complete objectives

- Interpret a derivative value as an instantaneous rate of change in context.
- State derivative units by dividing output units by input units.
- Use the sign and magnitude of a derivative to describe local behavior.
- Distinguish the meaning of a quantity, its first derivative, and its second derivative.

Strategic reminders

1. Read the notation before calculating: identify the dependent quantity, the independent variable, and their units.
2. Do not say that $Q'(a)$ is the amount of Q. It is the instantaneous rate of change of Q at input a.
3. A negative derivative describes direction of change; it does not imply the original quantity is negative.
4. For a table, $Q'(a)$ may be estimated using a symmetric difference when data on both sides are available.
5. Interpret $Q''(a)$ as the rate of change of Q'; include squared input units in the denominator.

Solved Example: Population interpretation

A city population $P(t)$ is measured in thousands of people, where t is years after 2025. Interpret $P'(4) = 2.7$.

Answer and Reasoning

In 2029, the city population is increasing at an instantaneous rate of 2.7 thousand people per year.

Lesson Mastery Practice

1. The mass $M(t)$ is in grams and t is in seconds. State the units of $M'(t)$ and $M''(t)$. Foundation

2. Revenue $R(q)$ is in dollars. Interpret $R'(120) = 38$. Core

3. Two quantities satisfy $A'(3) = -12$ and $B'(3) = 7$. Which is changing faster at 3? Explain. Advanced

4. Let $Q(t) = f(g(t))$. Write a contextual interpretation of $Q'(a) = f'(g(a))g'(a)$ in words. Challenge

AP-Style Reasoning Check

If $H(t)$ is height in meters and t is seconds, the units of $H''(t)$ are

(A) m

(B) m/s

(C) m/s^2

(D) s/m^2

Lesson 4.2: Straight Line Motion

Complete objectives

- Relate position, velocity, speed, and acceleration using derivatives.
- Determine direction of motion, rest times, and changes in speed.
- Analyze motion from formulas, graphs, and tables.
- Calculate displacement and total distance over intervals using turning times.

Strategic reminders

1. Find velocity and acceleration before solving sign questions.
2. A particle moves right when $v > 0$, left when $v < 0$, and is at rest when $v = 0$.
3. Use a sign chart for both v and a; do not infer speeding up from $a > 0$ alone.
4. To compute total distance without integration, locate all times where velocity changes sign and add absolute changes in position.
5. A zero velocity can be a stop without a direction reversal if the sign of velocity does not change.

Solved Example: Direction and rest

For $s(t) = t^3 - 6t^2 + 9t$, find the velocity and the times the particle is at rest.

Answer and Reasoning

$v(t) = 3t^2 - 12t + 9 = 3(t-1)(t-3)$. The particle is at rest at $t = 1$ and $t = 3$.

Lesson Mastery Practice

5. If $s(t) = 5t^2 - 3t + 1$, find $v(t)$ and $a(t)$. Foundation

6. For $s(t) = t^3 - 9t$, find all rest times for $t \geq 0$. Core

7. For $s(t) = t^3 - 6t^2 + 9t$, determine intervals of right and left motion for $t > 0$. Advanced

8. Construct a differentiable velocity function that is zero at $t = 1$ but does not reverse direction there. Challenge

AP-Style Reasoning Check

A particle is slowing down when

(A) $v > 0, a > 0$

(B) $v < 0, a < 0$

(C) v and a have opposite signs

(D) $a = 0$

Lesson 4.3: Other Rates of Change

Complete objectives

- Calculate and interpret derivatives in scientific, economic, and geometric contexts.
- Distinguish a modeled quantity from its instantaneous rate of change.
- Use correct units and signs in applied interpretations.
- Combine derivative rules to calculate rates for derived quantities.

Strategic reminders

1. Begin by naming the independent and dependent variables and their units.
2. A derivative can be computed symbolically, estimated from data, or read as slope from a graph; the interpretation format remains the same.
3. Do not confuse a large amount with a large rate. Compare rates using derivative values and magnitudes.
4. For marginal interpretations, say 'approximately' because a derivative predicts a small discrete change.
5. When a new quantity is built from several changing quantities, apply the appropriate product, quotient, or Chain Rule.

Solved Example: Marginal profit

Profit is $P(q) = 70q - 0.04q^2 - 5000$ dollars. Find and interpret $P'(600)$.

Answer and Reasoning

$P'(q) = 70 - 0.08q$, so $P'(600) = 22$. Near 600 units, one additional unit increases profit by approximately \$22.

Lesson Mastery Practice

9. The radius r is in cm and area A in cm^2. State units of dA/dr. Foundation

10. Volume is $V(r) = \frac{4}{3}\pi r^3$. Find $V'(5)$. Core

11. A rectangular region has $A(t) = L(t)W(t)$. If $L = 8$, $W = 5$, $L' = 0.3$, and $W' = -0.2$, find A'. Advanced

12. A quantity Q has units joules and depends on temperature T in kelvins. What are the units and meaning of d^2Q/dT^2? Challenge

AP-Style Reasoning Check

If $A(r)$ is area in m^2 and r is meters, units of $A'(r)$ are

(A) m^3

(B) m^2

(C) m

(D) 1/m

Lesson 4.4: Related Rates

Complete objectives

- Identify changing quantities linked by a geometric or physical constraint.
- Differentiate a constraint with respect to time using implicit differentiation.
- Substitute information from a specified instant and solve for an unknown rate.
- Interpret the sign and units of the resulting rate.

Strategic reminders

1. Draw and label a diagram; identify constants, variables, known rates, and the requested rate.
2. Write one constraint containing the target variable and the variables whose rates are known.
3. Differentiate every changing quantity with respect to the same independent variable.
4. Only after differentiation, substitute the values at the specified instant and solve.
5. State the final rate with sign, units, and a complete contextual interpretation.

Solved Example: Expanding circle

A circle's radius increases at 3 cm/s. How fast is its area increasing when $r = 5$ cm?

Answer and Reasoning

$A = \pi r^2$, so $A' = 2\pi r r' = 2\pi(5)(3) = 30\pi$ cm^2/s.

Lesson Mastery Practice

13. Differentiate $x^2 + y^2 = 25$ with respect to t. **Foundation**

14. A circle area increases at 12π cm^2/s. Find r' when $r = 3$ cm. **Core**

15. Air enters a sphere at 100 cm^3/s. Find r' when $r = 4$ cm. **Advanced**

16. A right circular cone preserves volume while its height increases. Derive the relation between r'/r and h'/h.
Challenge

AP-Style Reasoning Check

In a related-rates problem, numerical values should usually be substituted

(A) before writing a constraint
(B) before differentiating
(C) after differentiating
(D) only after solving the entire problem

Lesson 4.5: Linear Approximation

Complete objectives

- Construct the linearization of a differentiable function at a convenient input.
- Approximate nearby values using a tangent line.
- Use differentials to estimate changes and propagated error.
- Use concavity to determine whether a linear approximation overestimates or underestimates.

Strategic reminders

1. Choose a base point a where both $f(a)$ and $f'(a)$ are easy to compute.
2. Write the tangent line first; then substitute the nearby target input.
3. Keep exact arithmetic until the final approximation, especially for radicals and logarithms.
4. Use the sign of f'' on the interval between a and x to classify over- or underestimation.
5. For measurement error, use $|dy| \approx |f'(x)|\,|dx|$ as an estimate of maximum propagated error.

Solved Example: Square root

Use linearization at $a = 25$ to approximate $\sqrt{25.6}$.

Answer and Reasoning

$f(x) = \sqrt{x}$, $f(25) = 5$, $f'(25) = 1/10$. Thus $L(x) = 5 + (x - 25)/10$ and $\sqrt{25.6} \approx 5.06$.

Lesson Mastery Practice

17. Write the linearization of f at $x = a$. Foundation

18. Use $a = 1$ to approximate $\ln(1.04)$. Core

19. Find the linearization of $f(x) = \sin x$ at $a = \pi/6$. Advanced

20. Find a such that the linearization of e^x at a passes through the origin. Challenge

AP-Style Reasoning Check

The linearization of f at a is

(A) $f(a) + f'(x)(x - a)$

(B) $f(x) + f'(a)(x - a)$

(C) $f(a) + f'(a)(x - a)$

(D) $f'(a) + f(a)(x - a)$

Lesson 4.6: L'Hopital's Rule

Complete objectives

- Recognize the indeterminate quotient forms $0/0$ and ∞/∞.
- Verify the hypotheses of L'Hopital's Rule before differentiating numerator and denominator.
- Evaluate indeterminate limits, including repeated applications when justified.
- Rewrite selected products, differences, and powers into allowed quotient forms.

Strategic reminders

1. Always show the substitution that produces an allowed indeterminate form.
2. Differentiate the numerator and denominator as separate functions and then reevaluate the resulting limit.
3. A form such as $1/0$, $0/\infty$, or $\infty/0$ is not indeterminate and does not justify the rule.
4. For 1^{∞}, 0^{0}, or ∞^{0}, take logarithms, evaluate the logarithmic product, and exponentiate at the end.
5. State when repeated application is justified rather than applying derivatives mechanically.

Solved Example: One application

Evaluate $\lim_{x\to 0} \frac{e^{2x}-1}{x}$.

Answer and Reasoning

The form is $0/0$. Differentiate top and bottom: $\lim 2e^{2x} = 2$.

Lesson Mastery Practice

21. Identify the form of $\lim_{x\to 0} \frac{\sin x}{x}$. Foundation

22. Evaluate $\lim_{x\to 0} \frac{\ln(1+x)}{x}$. Core

23. Evaluate $\lim_{x\to 0} \frac{\tan x - x}{x^3}$. Advanced

24. Suppose $f(0) = g(0) = 0$, $g'(0) \neq 0$, and both are differentiable near 0. State a condition guaranteeing $\lim f/g = f'(0)/g'(0)$. Challenge

AP-Style Reasoning Check

L'Hopital's Rule may be applied directly when substitution gives

(A) $3/0$

(B) $0/5$

(C) $0/0$

(D) $\infty - \infty$

Unit 4 Difficult and Challenge Synthesis

1. At $t = 6$, a population satisfies $P = 1200$, $P' = -35$, and $P'' = 8$, where t is years. Interpret all three values precisely.
Hard

2. A particle has $s(t) = t^4 - 8t^3 + 18t^2$ on $0 \le t \le 6$. Determine all intervals of motion right/left and speeding up/slowing down.
Challenge

3. A 10-m ladder slides with its foot moving away from a wall at 1.2 m/s. Find the rate of change of the angle between ladder and ground when the foot is 6 m from the wall.
Difficult

4. Use differentials to estimate the maximum percentage error in the volume of a sphere when its radius is measured with at most 0.4% error. **Hard**

5. Evaluate $\lim_{x\to 0^+} x \ln x$ and $\lim_{x\to\infty} x\left(e^{1/x} - 1\right)$ using valid L'Hopital rewrites. **Difficult**

6. Evaluate $\lim_{x\to 0} (1 + 2x)^{3/x}$ by logarithmic transformation. **Challenge**

Cumulative AP-Style Checkpoint

1. If $A(t)$ is area in square meters and t is seconds, the units of $A''(t)$ are

(A) meters per second **(B)** square meters per second

(C) seconds squared per square meter **(D)** square meters per second squared

2. At an instant a particle has $v < 0$ and $a > 0$. The particle is

(A) moving right and slowing down **(B)** at rest

(C) moving left and slowing down **(D)** moving left and speeding up

3. If $C(q)$ is cost in dollars, then $C'(500) = 7.2$ means

(A) Exactly 500 additional units cost 7.20 **(B)** Near 500 units, producing one additional unit costs approximately 7.20

(C) The total cost of 500 units is 7.20 **(D)** Cost decreases by 7.20 at 500 units

4. For a circle with changing radius, which equation correctly relates area and radius rates?

(A) $A = 2\pi r \frac{dr}{dt}$ **(B)** $\frac{dA}{dr} = 2\pi \frac{dr}{dt}$

(C) $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$ **(D)** $\frac{dA}{dt} = \pi \left(\frac{dr}{dt}\right)^2$

5. If $f''(x) > 0$ near a, then the tangent-line approximation near a generally

(A) has no predictable error direction **(B)** underestimates $f(x)$

(C) overestimates $f(x)$ **(D)** is exact for every nearby input

6. Which limit is immediately eligible for L'Hopital's Rule after substitution?

(A) $\lim_{x\to 0} \frac{1}{x}$ **(B)** $\lim_{x\to\infty} \frac{1}{x^2}$

(C) $\lim_{x\to 0} \frac{x}{x^2+1}$ **(D)** $\lim_{x\to 0} \frac{e^x - 1}{x}$

7. If $C(t)$ is measured in dollars and t in months, what are the units of $C'(6)$?

(A) dollars **(B)** months per dollar

(C) dollars per month squared **(D)** dollars per month

8. A particle has $v(t) = (t-1)^2(t-3)$. At $t = 1$, the particle

(A) has a vertical tangent **(B)** is at rest but does not change direction

(C) changes from right to left **(D)** changes from left to right

9. A quantity Q is increasing at a decreasing rate at $t = a$. Which signs are correct?

(A) $Q'(a) > 0$ and $Q''(a) > 0$ **(B)** $Q'(a) < 0$ and $Q''(a) > 0$

(C) $Q'(a) > 0$ and $Q''(a) < 0$ **(D)** $Q'(a) < 0$ and $Q''(a) < 0$

10. A circle's radius increases at 2 cm/s. How fast is its area increasing when $r = 5$ cm?

(A) 20π cm/s **(B)** 21π cm^2/s

(C) 20π cm^2/s **(D)** 10π cm^2/s

11. Use $f(x) = \sqrt{x}$ linearized at $x = 36$ to approximate $\sqrt{36 + 0.01}$.

(A) $\frac{601}{100}$ **(B)** $\frac{7201}{1200}$

(C) $\frac{73}{12}$ **(D)** $\frac{7199}{1200}$

12. The standard first step for a limit of the form 1^∞ is to

(A) differentiate the base and exponent separately
(B) replace the expression by 1
(C) apply the Product Rule
(D) take the natural logarithm and analyze the resulting product or quotient

13. If $C(t)$ is measured in dollars and t in months, what are the units of $C'(6)$?

(A) dollars per month
(B) dollars per month squared
(C) dollars
(D) months per dollar

14. A particle's velocity is $v(t) = (t-3)(t-6)$. On which interval does it move left?

(A) $(-\infty, 3)$
(B) $(3, 6)$
(C) $(-\infty, 3) \cup (6, \infty)$
(D) $(6, \infty)$

15. A quantity Q is increasing at a decreasing rate at $t = a$. Which signs are correct?

(A) $Q'(a) < 0$ and $Q''(a) < 0$
(B) $Q'(a) > 0$ and $Q''(a) > 0$
(C) $Q'(a) > 0$ and $Q''(a) < 0$
(D) $Q'(a) < 0$ and $Q''(a) > 0$

16. A 10-m ladder has base $x = 6$ m moving away at 0.5 m/s. Find the rate of the top.

(A) 0.375 m/s
(B) −0.375 m/s
(C) 0.5 m/s
(D) −0.5 m/s

17. Use $f(x) = \sqrt{x}$ linearized at $x = 49$ to approximate $\sqrt{49 + 0.01}$.

(A) $\frac{701}{100}$
(B) $\frac{9799}{1400}$
(C) $\frac{9801}{1400}$
(D) $\frac{99}{14}$

18. Evaluate $\lim_{x\to 0} \dfrac{e^{2x}-1}{x}$.

(A) 1
(B) 0
(C) e^2
(D) 2

19. If $C(t)$ is measured in dollars and t in months, what are the units of $C'(6)$?

(A) dollars
(B) dollars per month
(C) dollars per month squared
(D) months per dollar

20. A particle's velocity is $v(t) = (t-2)(t-4)$. On which interval does it move left?

(A) $(4, \infty)$
(B) $(2, 4)$
(C) $(-\infty, 2)$
(D) $(-\infty, 2) \cup (4, \infty)$

Cumulative Free-Response Synthesis

FRQ1 – Complete motion analysis

A particle moves on a line with velocity $v(t) = t^3 - 6t^2 + 8t$ for $0 \le t \le 6$, and $s(0) = 4$.

(a) Find all times when the particle changes direction and state the direction on each interval. [4 points]

(b) Find acceleration and determine the intervals on which speed is increasing. [4 points]

(c) Interpret $v'(3)$ in context. [2 points]

FRQ2 – Related rates with a ladder

A 10-meter ladder leans against a wall. Its foot moves away from the wall at 1.2 m/s.

(a) Find the rate at which the top moves downward when the foot is 6 m from the wall. [3 points]

(b) Find the rate of change of the angle between the ladder and the ground at the same instant. [3 points]

(c) Explain why the two negative rates have different units and meanings. [2 points]

FRQ3 – Contextual first and second derivatives

The mass $M(t)$ of a sample is measured in grams, where t is minutes. At $t = 8$, $M = 42$, $M' = -1.6$, and $M'' = 0.25$.

(a) Interpret all three values, including units. [4 points]

(b) Is the mass loss becoming faster or slower near $t = 8$? Explain. [2 points]

(c) Use a local linear model to estimate $M(8.3)$. [2 points]